

The Algebra of the Chessboard: the Symmetry Group and Orbit Censuses

Theorem monograph T01 of the chess-dynamics-lab program

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Keywords: chessboard, D_4 group, orbits, Burnside lemma

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Setting

The chessboard 8×8 is the set of squares $S = \{0, \dots, 7\}^2$, identified with (file, rank) pairs. Every “particle physics” that we build on the board rests on the geometry of this set, and before defining fields, flows and energies one must know exactly the symmetry group of the board and its action on the squares. That group determines which quantities are invariant, which fields are equivariant, and which configurations count as the same up to rotation and reflection.

In the parent monograph “The Dynamic Principle” (the hodge-laboratory program) the μ_4 -equivariance of the grid was the key instrument of Part XV. Here we obtain the complete finite-group framework of the chessboard: the group D_4 of all eight symmetries of the square, its colour-preserving subgroup V_4 , and the exact orbit censuses of the action on the 64 squares. All censuses are verified by check C1 of the dynamics.py laboratory and coincide with the direct computation, which makes the theorem simultaneously a definition and a correctness certificate of the whole geometric layer of the project.

We stress the role of the colouring: the 32 white and 32 black squares define the parity homomorphism $\chi(f, r) = (f + r) \bmod 2$. Colour preservation is exactly the preservation of χ , and the subgroup we single out is the kernel of this difference action. For piece-particles this is essential: a bishop lives on one colour only, a knight changes colour with every move, and a queen move mixes both colours.

Theorem

Theorem T01 (orbit censuses of the chessboard). The symmetry group of the board is D_4 , $|D_4| = 8$. The colour-preserving subgroup is $V_4 = \{\text{id}, \rho_{180}, \sigma_{d1}, \sigma_{d2}\} \cong$

$\mathbb{Z}_2 \times \mathbb{Z}_2$. The orbits of the V_4 -action on the 64 squares: exactly 20 orbits — 8 diagonal orbits of length 2 and 12 off-diagonal orbits of length 4. The orbits of the D_4 -action: exactly 10. In both cases the census matches the Burnside evaluation.

Proof

Every symmetry of the square maps the set of squares to itself and is determined by the image of one corner square; hence there are exactly eight such symmetries: the identity, the rotations by 90° , 180° , 270° and four reflections (two midlines and two diagonals). They form the dihedral group D_4 of order 8. In (file f , rank r) coordinates, each in $0..7$: the rotation by 90° is $(f, r) \mapsto (r, 7 - f)$, the rotation by 180° is $(f, r) \mapsto (7 - f, 7 - r)$, the reflection in the main diagonal is $\sigma_{d1}(f, r) = (r, f)$, and in the anti-diagonal $\sigma_{d2}(f, r) = (7 - r, 7 - f)$.

We verify the preservation of the parity $\chi(f, r) = (f + r) \bmod 2$ element by element. The identity preserves χ trivially. For ρ_{180} the coordinate sum of the image is $(7 - f) + (7 - r) = 14 - (f + r)$, congruent to $f + r$ modulo 2 because 14 is even. For σ_{d1} the sum does not change at all: $r + f = f + r$. For σ_{d2} the image sum is $(7 - r) + (7 - f) = 14 - (f + r)$ — the same parity again. Hence all four elements preserve the colour; they are closed under composition (each is an involution, and the product of any two yields the third), so they form a subgroup isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

The remaining four elements break the colour. For the rotation by 90° the image sum is $r + (7 - f) = 7 + (r - f)$, and $\chi(f, r) + \chi(\rho_{90}(f, r)) = (f + r) + (r - f) + 7 = 2r + 7$ is odd, so the parity always flips. The same holds for ρ_{270} ; for the midline reflections $(7 - f) + r = 7 + (r - f)$ flips the parity because 7 is odd. Consequently the colour-preserving subgroup is exactly the selected quartet, and V_4 is indeed isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Now the orbits of V_4 . A square of the main diagonal $f = r$ is fixed by σ_{d1} , so its stabilizer is non-trivial, of order 2, and the orbit length is $|V_4|/2 = 2$: the orbit of (r, r) consists of the square itself and the opposite square $(7 - r, 7 - r)$. The main diagonal holds 8 squares, split into pairs, giving exactly 4 orbits of length 2. Likewise the 8 squares of the anti-diagonal $f + r = 7$ are fixed by σ_{d2} and give 4 more orbits of length 2. The diagonals do not intersect (the system $f = r$, $f + r = 7$ gives $2f = 7$, which has no integral solution), so all 8 orbits are distinct. The remaining $64 - 16 = 48$ squares lie on neither diagonal; for them the stabilizer is trivial (each diagonal involution moves the square), the orbit length is 4, and we get exactly $48/4 = 12$ orbits of length 4. In total $8 + 12 = 20$ orbits, and the length count $8 \cdot 2 + 12 \cdot 4 = 64$ adds up.

For the full group D_4 we apply the Burnside lemma. The numbers of fixed squares: the identity fixes 64; the rotations by 90° and 270° fix none, because the equation $(f, r) = (r, 7 - f)$ forces $f = r = 7 - f$, i.e. $2f = 7$, which has no integral solution; the rotation by 180° fixes none for the same reason; the midline reflections fix none (the equation $f = 7 - f$ is unsolvable over the integers); each diagonal fixes 8 squares. Altogether $|\text{Orbits}(D_4)| = (64 + 0 + 0 + 0 + 0 + 0 + 8 + 8)/8 = 80/8 = 10$. The same lemma applied to V_4 gives $(64 + 0 + 8 + 8)/4 = 20$ — a complete agreement with the direct orbit count. The theorem is proved.

Protocol data

Orbit censuses (check C1 of dynamics.py / the polyglot core)

Quantity	Value
$ D_4 $	8
$ V_4 $	4
V_4 orbits (direct count)	20
V_4 orbits (Burnside)	$(64 + 0 + 8 + 8)/4 = 20$
Orbits of length 2 (diagonals)	8
Orbits of length 4 (off-diagonal)	12
D_4 orbits (direct count)	10
D_4 orbits (Burnside)	$(64 + 0+0+0+0+0+8+8)/8 = 10$

Summary of what is proved

We proved the complete orbit framework of the chessboard: the colour-preserving subgroup $V_4 \cong \mathbb{Z}_2^2$ is isolated by a component-wise parity check, the census of its orbits is 20 with the structure $8 + 12$, the census of the orbits of the full group D_4 is 10, and both censuses are confirmed by the Burnside lemma. The proof mechanism is finite-group-theoretic (stabilizers, orbit lengths, Burnside), and it serves as the template for all subsequent equivariant claims of the program: the threat fields (T04) inherit the D_4 -symmetry exactly to the extent the move rules of the pieces inherit it.

Verification

The check reproduces with a single command: `python3 dynamics.py -run C1` (the laboratory) or `python3 polyglot/python/chess_core.py` (the polyglot core, the C1 line of the table). The direct orbit count is implemented as an exhaustive pass over the group action, the Burnside estimate as a sum of fixed points; the agreement of both numbers with the expected 20 and 10 yields the PASS verdict. The same core is implemented in C, Rust, Go, Julia, JavaScript and Java, and in all seven implementations the C1 line must coincide bit for bit.